ENERGIA: USB-C, carga de bateria, reparto de carga y regulador 3.3 V

J1 — USB-C 16P
VBUS (A4, B9), CC1 (A5), CC2 (B5), USB_DM (A7, B7), USB_DP (A6, B6), SBU1 (A8), SBU2 (B8), SHIELD, GND (A1, B12, S1)

U1 — TP4056
VBUS, CE, CHRG_N, STDBY_N, Vcc, BAT, TEMP, PROG, EPAD, GND
VBAT, PROG

J2 — BAT+, BAT−
Bateria LiPo soldada a pads.
Usar una con proteccion integrada

Q1 — AO3401A
VBUS, VBAT, VSYS

U2 — AP2112K-3.3
VSYS, PWR_EN, VIN, EN, VOUT, GND
3V3_REG

D3 — B5819W
VSYS, VBUS

SW1 — ENCENDIDO
PWR_EN, VSYS, GND

R1 5.1k CC1 · R2 5.1k CC2 · C1 10uF VBUS · R3 3k PROG · C2 10uF VBAT · R6 100k VBUS · C3 1uF VSYS · C4 10uF 3V3_REG · R15 100k PWR_EN · R11 1M VBAT · R12 1M BAT_SENSE · C11 100nF BAT_SENSE · R4 1k VBUS · R5 1k VBUS
R11 → BAT_SENSE; R4 → LED_CHG_A; R5 → LED_FULL_A

JP1 — 3V3_REG, +3V3, 3V3

D1 — CHRG_N, LED_CHG_A — CARGANDO (rojo)
D2 — STDBY_N, LED_FULL_A — CARGA COMPLETA (verde)

Puntos de prueba y banderas de alimentacion
TP1 VBUS · TP2 VBAT · TP3 VSYS · TP4 +3V3 · TP5 GND · TP6 EN · TP7 BTN_MENU · TP8 UART_TX · TP9 UART_RX
PWR_FLAG VBUS · PWR_FLAG VSYS · PWR_FLAG +3V3 · PWR_FLAG GND

PANTALLA (conector PROVISIONAL)

J3 — LCD 2.4in ST7789V
+3V3, GND, LCD_SCK, LCD_MOSI, LCD_CS, LCD_DC, LCD_RST, LCD_LEDA, LCD_LEDK

Q2 — AO3400A
LCD_BL, LCD_LEDK, GND

R13 10R (por definir) +3V3 → LCD_LEDA
R14 100k LCD_BL → GND

Pines y resistencia de luz de fondo:
se definen al tener el panel en la mano

microSD (SPI)

J4 — TF-01A
GND, GND, SD_MISO, DAT0(DO), DAT1(RSV), DAT2(RSV), CDDAT3(CS), CMD(DI), CLK(SCLK), VDD, CD, VSS, GND, GND, +3V3, GND

R8 10k +3V3 → SD_CS · R9 10k +3V3 → SD_MISO · C8 100nF +3V3 → GND · C12 10uF +3V3 → GND

ESP32-S3-WROOM-1-N16R8

U3 — ESP32-S3-WROOM-1-N16R8
+3V3, EN, 3V3, GND, PSRAM
BTN_MENU 27, BAT_SENSE 38, BTN_L 38, BTN_X 15, BTN_UP 4, BTN_DOWN 5, BTN_LEFT 6, BTN_RIGHT 7, BTN_A 12, LCD_DC 17, LCD_CS 18, LCD_MOSI 19, LCD_SCK 20, LCD_BL 21, LCD_RST 22, SD_MOSI 8, SD_SCK 9
TXD0 37 UART_TX, RXD0 36 UART_RX, IO17 10 SD_MISO, IO18 11 SD_CS, USB_D- 13 USB_DM, USB_D+ 14 USB_DP, IO21 23 BTN_B, IO35 28, IO36 29, IO37 30, IO38 31 BTN_R, IO39 32 BTN_OPTION, IO40 33 I2S_BCLK, IO41 34 I2S_LRCLK, IO42 35 I2S_DIN, IO45 26, IO46 16 BTN_Y, IO47 24 BTN_START, IO48 25 BTN_SELECT

SW2 — RESET
EN

R7 10k +3V3 → EN · C5 1uF EN → GND · C6 10uF +3V3 → GND · C7 100nF +3V3 → GND

BOTONES (cada uno de su GPIO a GND, pull-up interno)

SW3 UP — BTN_UP
SW4 DOWN — BTN_DOWN
SW5 LEFT — BTN_LEFT
SW6 RIGHT — BTN_RIGHT
SW7 A — BTN_A
SW8 B — BTN_B
SW9 X — BTN_X
SW10 Y — BTN_Y
SW11 START — BTN_START
SW12 SELECT — BTN_SELECT
SW13 MENU — BTN_MENU
SW14 OPTION — BTN_OPTION
SW15 L (gatillo lateral) — BTN_L
SW16 R (gatillo lateral) — BTN_R

MENU (GPIO 0): presionado al encender entra en modo de grabacion.
Botones frontales: solo se usan dos pads en diagonal. L y R son de actuacion lateral.

AUDIO (I2S)

U4 — MAX98357A
I2S_DIN, I2S_BCLK, I2S_LRCLK, AMP_SD, VSYS, VDD, DIN, BCLK, LRCLK, SD_MODE, GAIN_SLOT, GND, EP PAD, OUTP, OUTN
SPK_P, SPK_N

J5 — BOCINA+, BOCINA−
SPK_P, SPK_N

R10 1M AMP_SD → VSYS · C9 10uF VSYS → GND · C10 100nF VSYS → GND

GAIN sin conectar = 9 dB

Borrador generado. Sin probar en hardware.

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Title: MagSafe Console	
Size: A2	Date:
KiCad E.D.A. 9.0.9	**Rev: 0.1**
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